

Counterexample Packet

The Gaussian–triangle product is not extremal

Status: candidate counterexample, likely valid

Scope: Question 3, second clause; the first clause remains open

Source and original question

The source is Jamning–Matolcsi–Révész, *On the extremal rays of the cone of positive, positive definite functions*, arXiv:0801.0941, published in the Journal of Fourier Analysis and Applications [1]. Question 3 occurs in the Conclusion on PDF page 18.

— Finally, all the extremals we found, except the gaussians, either have at least a zero, or their Fourier transform has at least a zero. Borisov [Bor] conjectured that the only extremals in $L^2(\mathbb{R}) \cap \mathcal{C}(\mathbb{R})$ that are strictly positive and that have strictly positive Fourier transform are the gaussian functions $e^{-a\pi x^2}$, $a > 0$. We are inclined to believe that this is false. This is linked to Question 1 above which leads us to the following more precise question:

Question 3.

Consider the two functions $f = \chi_{[-1/2, 1/2]} * \chi_{[-1/2, 1/2]}$ and $\gamma = e^{-\pi x^2}$, and recall that they are extremal positive positive definite functions in Ω_1^+ . Is $f * \gamma$ extremal in Ω_1^+ and, if so, is $\gamma(f * \gamma)$ extremal in Ω_1^+ ?

A positive answer to the second question would of course provide a counterexample to Borisov’s conjecture.

Writing

$$f = \mathbf{1}_{[-1/2, 1/2]} * \mathbf{1}_{[-1/2, 1/2]} = (1 - |x|)_+, \quad \gamma(x) = e^{-\pi x^2},$$

the source asks whether $f * \gamma$ is extremal in Ω_1^+ and, if so, whether

$$G(x) := \gamma(x)(f * \gamma)(x)$$

is extremal. The result below gives a negative answer for G .

Definitions and Fourier convention

We use

$$\widehat{v}(\xi) = \int_{\mathbb{R}} v(x) e^{-2\pi i x \xi} dx.$$

The cone Ω_1^+ consists of the continuous even functions on $\mathbb{R}$ that are both nonnegative and positive definite. For the integrable functions used below, nonnegativity of the Fourier transform implies positive definiteness by Bochner’s theorem and Fourier inversion [2, 1].

A nonzero $V \in \Omega_1^+$ is an extremal ray generator if every decomposition $V = V_1 + V_2$ with $V_1, V_2 \in \Omega_1^+$ has $V_j = c_j V$ for nonnegative constants c_j .

Proof intuition

Completing the square moves half of the Gaussian from the pointwise product into a convolution:

$$G = 2q * \gamma_2, \quad q(x) = (1 - 2|x|)_+ e^{-2\pi x^2}, \quad \gamma_2(x) = e^{-2\pi x^2}.$$

The compactly supported factor q lies in Ω_1^+ and has a strictly positive Fourier transform Q . The jumps of q' at $-1/2, 0, 1/2$ show that Q is bounded below by a positive multiple of $(1 + \xi^2)^{-1}$. Multiplying q by x^2 produces a perturbation whose Fourier transform has the same upper decay order. Consequently, for sufficiently small $\varepsilon > 0$, both $q(1 + \varepsilon x^2)$ and $q(1 - \varepsilon x^2)$ remain nonnegative on both the time and Fourier sides. Convolution of these two nonproportional functions with γ_2 gives a nontrivial decomposition of G in Ω_1^+ .

Counterexample theorem

Theorem 1. *Let $f = (1 - |x|)_+$ and $\gamma(x) = e^{-\pi x^2}$. Then*

$$G(x) = \gamma(x)(f * \gamma)(x)$$

does not generate an extremal ray of Ω_1^+ .

More precisely, with

$$q(x) = (1 - 2|x|)_+ e^{-2\pi x^2}, \quad \gamma_2(x) = e^{-2\pi x^2},$$

there is an $\varepsilon > 0$ such that

$$q_{\pm}(x) = q(x)(1 \pm \varepsilon x^2)$$

both belong to Ω_1^+ , and

$$G = (q_+ * \gamma_2) + (q_- * \gamma_2)$$

is a decomposition into two nonproportional elements of Ω_1^+ .

Proof. 1. Gaussian refactorization. Since f is supported on $[-1, 1]$,

$$\begin{aligned} G(x) &= e^{-\pi x^2} \int_{-1}^1 f(t) e^{-\pi(x-t)^2} dt \\ &= \int_{-1}^1 f(t) e^{-2\pi(x-t/2)^2} e^{-\pi t^2/2} dt \\ &= 2 \int_{-1/2}^{1/2} f(2s) e^{-2\pi s^2} e^{-2\pi(x-s)^2} ds \\ &= 2(q * \gamma_2)(x). \end{aligned}$$

2. Strict positivity of $\hat{q}$. Put $h(x) = (1 - 2|x|)_+ = f(2x)$. Both h and γ_2 belong to Ω_1^+ , and $q = h\gamma_2$; hence $q \in \Omega_1^+$. Moreover,

$$\hat{h}(\xi) = \frac{1}{2} \left(\frac{\sin(\pi\xi/2)}{\pi\xi/2} \right)^2 \geq 0, \quad \hat{\gamma}_2(\xi) = \frac{1}{\sqrt{2}} e^{-\pi\xi^2/2} > 0.$$

Thus, if $Q = \widehat{q}$, then

$$Q = \widehat{h} * \widehat{\gamma}_2 > 0 \quad \text{on } \mathbb{R},$$

because the Gaussian factor is everywhere positive and $\widehat{h}$ is nonnegative and nonzero away from only a discrete set.

3. A global lower bound for Q . Extend q by zero outside $[-1/2, 1/2]$. Its derivative has jumps

$$[q']_{-1/2} = 2e^{-\pi/2}, \quad [q']_0 = -4, \quad [q']_{1/2} = 2e^{-\pi/2}.$$

Consequently, in the sense of distributions,

$$q'' = r - 4\delta_0 + 2e^{-\pi/2}(\delta_{-1/2} + \delta_{1/2}),$$

where $r \in L^1(\mathbb{R})$ is the ordinary second derivative on the smooth pieces. Taking Fourier transforms gives, for $\xi \neq 0$,

$$-4\pi^2\xi^2Q(\xi) = \widehat{r}(\xi) - 4 + 4e^{-\pi/2}\cos(\pi\xi).$$

The Riemann–Lebesgue lemma yields

$$Q(\xi) = \frac{1 - e^{-\pi/2}\cos(\pi\xi)}{\pi^2\xi^2} + o(\xi^{-2}) \quad (|\xi| \rightarrow \infty).$$

Since $1 - e^{-\pi/2}\cos(\pi\xi) \geq 1 - e^{-\pi/2} > 0$, this gives a positive lower bound $Q(\xi) \geq c_\infty/\xi^2$ outside a compact interval. On that compact interval, strict positivity and continuity of Q give another positive lower bound. Therefore there is $c > 0$ such that

$$Q(\xi) \geq \frac{c}{1 + \xi^2} \quad (\xi \in \mathbb{R}). \quad (1)$$

4. A matched perturbation bound. Let $u(x) = x^2q(x)$ and $U = \widehat{u}$. The function u is continuous, compactly supported, piecewise smooth, and its distributional second derivative is a finite signed measure. Hence

$$|U(\xi)| \leq \frac{C}{1 + \xi^2} \quad (\xi \in \mathbb{R}) \quad (2)$$

for some $C > 0$: for large $|\xi|$ this follows from $-4\pi^2\xi^2U(\xi) = \widehat{u''}(\xi)$, and boundedness on a compact interval follows from $u \in L^1$.

Choose

$$0 < \varepsilon < \min \left\{ 4, \frac{c}{2C} \right\}.$$

Because $|x| \leq 1/2$ on the support of q , $q_\pm = q(1 \pm \varepsilon x^2)$ are nonnegative. Equations (1)–(2) give

$$\widehat{q}_\pm(\xi) = Q(\xi) \pm \varepsilon U(\xi) \geq Q(\xi) - \varepsilon |U(\xi)| \geq \frac{c/2}{1 + \xi^2} > 0.$$

The functions $q_\pm$ are continuous and integrable, and their Fourier transforms are nonnegative (indeed integrable, by the same piecewise-smooth decay argument). Therefore $q_\pm \in \Omega_1^+$.

5. Nontrivial decomposition of G . Since $q_+ + q_- = 2q$, Step 1 gives

$$(q_+ * \gamma_2) + (q_- * \gamma_2) = 2q * \gamma_2 = G.$$

Each summand belongs to Ω_1^+ . They are not proportional to G : if, for example, $q_+ * \gamma_2 = \lambda G$, taking Fourier transforms and cancelling the nowhere-zero transform of γ_2 would give $q_+ = 2\lambda q$, contrary to $q_+/q = 1 + \varepsilon x^2$ on $(-1/2, 1/2)$. Hence the displayed decomposition is nontrivial, and G is not extremal. $\square$

Verification report

Step	Independent check
Refactorization	Expanding $2(x - t/2)^2 + t^2/2$ gives $x^2 + (x - t)^2$ exactly; the substitution $t = 2s$ supplies the factor 2.
Fourier strict positivity	$q = h\gamma_2$ makes Q a convolution of a nonnegative sinc-square with a strictly positive Gaussian.
Jump asymptotic	Direct one-sided differentiation gives jumps $2e^{-\pi/2}, -4, 2e^{-\pi/2}$; the resulting leading bracket is uniformly positive because $e^{-\pi/2} < 1$.
Perturbation	u'' is a finite measure, so $U = O(\xi^{-2})$, matching the proved lower order of Q . A single small ε works globally.
Nonproportionality	Gaussian convolution is injective because its Fourier transform never vanishes.

No computational step is used in the proof. The auxiliary script `code/numerical_sanity_check.py` checks the exact refactorization and samples $Q \pm \widehat{x^2 q}$ on $0 \leq \xi \leq 100$ for $\varepsilon = 1$. Finite sampling is recorded only as a sanity check and is not evidence for the global inequalities.

Scope and novelty check

This theorem completely answers the second clause of Question 3 in the negative. It does not decide the first clause, namely whether $f * \gamma$ is extremal. It also shows that the function proposed in the source cannot serve as a counterexample to Borisov’s conjecture, because it is not extremal; it does not decide that broader conjecture.

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:0801.0941, the source title, “extremal rays,” and the Gaussian–triangle expressions. No prior packet or attempt was found. External searches used the exact Question 3 phrases, the source title and DOI, “product of two extremals,” “positive positive-definite,” “Borisov’s conjecture,” and arXiv-restricted variants. They located the source paper and general later work on Fourier positivity, but no paper stating an answer to either exact clause. Novelty confidence is therefore moderate rather than absolute; citation databases and non-indexed literature were not exhaustively searched.

Human review recommendation

Send for mathematical review. The highest-value checks are the distributional jump computation in Step 3, the global lower-bound passage, and the assertion that u'' is a finite measure in Step 4. The construction is otherwise elementary and has no unproved dependency.

References

- [1] P. Jaming, M. Matolcsi, and S. Gy. Révész, *On the extremal rays of the cone of positive, positive definite functions*, J. Fourier Anal. Appl. 15 (2009), 561–582; arXiv:0801.0941, doi:10.1007/s00041-008-9057-6.
- [2] S. Bochner, *Monotone Funktionen, Stieltjessche Integrale und harmonische Analyse*, Math. Ann. 108 (1933), 378–410.